

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

FORESIGHT DIAGNOSTICS, INC.,
Petitioner,

v.

PERSONALIS, INC.,
Patent Owner.

IPR2023-00317
Patent 11,408,033 B2

Before ULRIKE W. JENKS, ROBERT A. POLLOCK, and
TIMOTHY G. MAJORS, *Administrative Patent Judges*.

POLLOCK, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable

35 U.S.C. § 318(a)

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review challenging the patentability of claims 1–23 of U.S. Patent No. 11,408,033 B2 (“the ’033 patent,” Ex. 1001). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e) (2018). Having reviewed the parties’ arguments and cited evidence, for the reasons discussed below, we find that Petitioner has demonstrated by a preponderance of the evidence that claims 1–23 are unpatentable for the reasons set forth in the Petition.

A. Procedural History

Foresight Diagnostics Inc. (“Petitioner” or “Foresight”) filed a Petition for an *inter partes* review of claims 1–23 of the ’394 patent. Paper 1 (“Pet.”). Personalis, Inc. (“Patent Owner” or “Personalis”) timely filed a Preliminary Response. Paper 7. With our authorization (*see* Ex. 3001), Petitioner filed a Reply to the Preliminary Response (Paper 8); Patent Owner filed a corresponding Sur-reply (Paper 10). In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 11 (“Institution Decision” or “DI”).

After institution, Patent Owner filed a Response. Paper 18 (“POR”). Petitioner filed a Reply. Paper 41 (“Reply”). Patent Owner filed a Sur-reply. Paper 38 (“Sur-reply”). With our authorization, Petitioner further filed a Response to Patent Owner’s Sur-reply. Paper 42 (“Resp.”).

On March 19, 2024, we held an oral hearing, the transcript of which is of record. Paper 46 (“Tr.”).

B. Real Parties-in-Interest

Petitioner identifies itself, Foresight Diagnostics Inc., as the real party-in-interest. Paper 32, 1. Patent Owner also identifies only itself, Personalis, Inc., as the real party-in-interest in this proceeding. Paper 6, 1.

C. Related Matters

In addition to the '033 patent at issue here, Petitioner concurrently challenges claims of related U.S. Patent No. 11, 384,394 B2 (“the '394 patent”) in IPR2023-00224. The '033 and '394 patents share substantially the same disclosure.

The parties further identify as “related proceedings,” Petitioner’s challenge of Personalis’s U.S. Patent Nos. 11,299,783 (“the '783 patent”) and 10,450,611 (“the '611 patent”) in IPR2023-00545 and IPR2023-00546, respectively. Paper 6, 1; Paper 33, 2. IPR2024-00170 involving U.S. Patent No. 11,584,968 (“the '968 patent”) is also before us. *See* Paper 33, 2

The '394, '033, '738, and '611 patents are at issue in *Personalis, Inc. v. Foresight Diagnostics, Inc.*, C.A. No. 1:22-cv-01913 (D. Colo.) (“the 01913 litigation”). *See* Pet. 3; Paper 6, 1; Ex. 1010, 1, 3–6; Exhibit 1010, 1 (First Amended Complaint adding '033 patent). Petitioner further identifies as a “related matter[,]” *Personalis, Inc. v. Foresight Inc.*, 1:23-cv-01623 (D. Colo.), which further involves the '968 patent. Paper 33, 2; Ex. 1010, 1; IPR2024-00170, Paper 1, 1, 3 (stating that Foresight has moved to consolidate the two district court proceedings).

D. Asserted Challenges to Patentability

Petitioner challenges the patentability of claims 1–23 on the following bases: (Pet. 16):

Ground	Claims Challenged	35 U.S.C. §	Reference(s)/Basis
1	1–7, 10–23	§ 102	Forsheew ¹
2	1–23	§ 103	Forsheew, Wagle ²
3	1–9, 11–23	§ 103	Wagle, Chan ³

In support of its patentability challenge, Petitioner relies on, *inter alia*, the First and Second Declarations of John Quackenbush, Ph.D., Exhibits 1020 and 1225, respectively. Patent Owner relies on, *inter alia*, the Declaration of Henry Morrice Furneaux, Ph.D., Exhibit 2031. Patent Owner further relies on the Declarations of Jonathan MacQuitty (Ex. 2033), Doug Zeman (Ex. 2034), Dan Norton (Ex. 2035), and John West (Ex. 2032).⁴

E. The '033 Patent and Related Background

1. Priority

The '033 patent, titled Methods and Systems for Genetic Analysis, issued to Bartha et al., from U.S. Application 17/078,857, filed October 23,

¹ Tim Forsheew, et al., “Noninvasive Identification and Monitoring of Cancer Mutations by Targeted Deep Sequencing of Plasma DNA,” 4 *Science Translational Medicine* 136ra68 (2012). Ex. 1030, Ex. 1032 (Supplementary Materials).

² Nikhil Wagle et al., “High-Throughput Detection of Actionable Genomic Alterations in Clinical Tumor Samples by Targeted, Massively Parallel Sequencing,” Jan. 2012 *Cancer Discovery* 83–93. Ex. 1033.

³ K.C. Allen Chan *et al.*, “Cancer Genome Scanning in Plasma: Detection of Tumor-Associated Copy Number Aberrations, Single-Nucleotide Variants, and Tumoral Heterogeneity by Massively Parallel Sequencing,” 59(1) *Clin. Chem.* 211–224 (2013). Ex. 1008.

⁴ Mr. West is a named inventor of the challenged patent. Ex. 1001, code (72). Mr. West co-founded Personalis and served as its Chief Executive Officer from August 2011 through December 2022. Ex. 2023 ¶ 3.

2020, via a series of continuation and divisional applications first filed on December 27, 2013, and further claims benefit of priority to U.S. Provisional Application No. 61/753,828, filed on January 17, 2013 (“the ’828 Provisional Application”). Ex. 1001, code (12), (21), (22), (54), and (60).⁵ The parties dispute whether the ’033 patent is entitled to the benefit of the provisional filing date. *See, e.g.*, Reply 12–13; POR 2, 6.

2. Abstract and Specification

The Abstract of the ’033 patent broadly describes “systems and methods for sample processing and data analysis,” which can include “nucleic acid sample processing and subsequent sequencing.” Ex. 1001, code (57). According to the Abstract, the resultant “sequence information may be analyzed with the aid of a computer processor, and the analyzed sequence information may be stored in an electronic storage location that may include a pool or collection of sequence information and analyzed sequence information generated from the nucleic acid sample.” *Id.*

The ’033 Specification more specifically discloses methods for “predicting, diagnosing, and/or prognosing a status or outcome of a disease or condition in a subject.” *Id.* at 55:38–50. In some embodiments, this involves the collection and analysis of sequence information derived from body fluid or tissue, including “from a subject suffering from a cancer.” *Id.* at 38:57–67, 55:51–57:8.

⁵ The ’394 patent likewise issued from a series of continuation and division applications first filed on December 27, 2013, and claims benefit of priority to the ’828 Provisional Application such that the ’033 and ’394 patents share the same disclosure. As such, the parties’ declarants generally cite only one of the two substantially identical disclosures. *See, e.g.*, Ex. 2031 ¶ 57 n.3; Ex. 1225 ¶ 39.

3. Challenged Claims

Petitioner challenges claims 1–23, of which only claim 1 is independent and recites:

1. A method for analyzing a nucleic acid sample obtained from an individual, comprising:

(a) producing, with the aid of a computer processor, a plurality of capture probes, wherein said plurality of capture probes hybridize to a plurality of polymorphisms, wherein said plurality of polymorphisms are in sequences encoding genes with known biomedically interpretable variants, and wherein said plurality of polymorphisms are based on or extracted from one or more databases of polymorphisms and are observed in a population of one or more samples;

(b) contacting said nucleic acid sample with said plurality of capture probes produced in (a);

(c) conducting a sequencing assay on a subset of nucleic acid molecules generated from said nucleic acid sample in step (b) to yield a result comprising a nucleic acid sequence, thereby analyzing said nucleic acid sample; and

(d) repeating steps (b)–(c) on a subsequently obtained biological sample from said individual.

Ex. 1001, 79:46–65. Depending from claim 1, claim 2 entails “generating a biomedical report that includes biomedical information” referenced in claim step 1(a), whereas dependent claim 3 recites that this biomedical information

is predictive, prognostic, or diagnostic of one or more biomedical features selected from the group consisting of disease state, efficacy of a drug therapy, prediction of optimal drug dosage, recommendation of one or more therapies, and recommendation of a course of treatment of a disease.

Id. at 79:66–80:32. Also depending from claim 1, claim 10 specifies that the “capture probes further comprise barcodes.” *Id.* at 80:50–51.

II. ANALYSIS

A. Legal Standards

“In an IPR, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

Petitioner contends that claims 1–7 and 10–23 are anticipated by the prior art. Pet. 11. To show anticipation under 35 U.S.C. § 102, each and every claim element, arranged as in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). The prior art need not, however, use the same words as the claims to find anticipation. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009). In evaluating anticipation, it is permissible to take into account not only the literal teachings of the prior art reference, but also the inferences the skilled artisan would draw from it. *Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. At Harbor-UCLA Med. Ctr.*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017) (holding that the “dispositive question regarding anticipation is whether one skilled in the art would reasonably understand or infer from a prior art reference that every claim element is disclosed in that reference”). As such, “a reference can anticipate a claim even if it does not expressly spell out all the limitations arranged or combined as in the claim, if a person of skill in the art, reading the reference, would at once envisage

the claimed arrangement or combination.” *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d 1376, 1381 (Fed. Cir. 2015) (internal quotation and alterations omitted).

Petitioner also contends that each claim of the ’033 patent is unpatentable as obvious under § 103(a). The Supreme Court in *KSR International Co. v. Teleflex Inc.*, 550 U.S. 398 (2007), reaffirmed the framework for determining obviousness set forth in *Graham v. John Deere Co.*, 383 U.S. 1 (1966). The *KSR* Court summarized the four factual inquiries set forth in *Graham* (383 U.S. at 17–18) that are applied in determining whether a claim is unpatentable as obvious under 35 U.S.C. § 103 as follows: (1) determining the scope and content of the prior art; (2) ascertaining the differences between the prior art and the claims at issue; (3) resolving the level of ordinary skill in the art; and (4) considering objective evidence indicating obviousness or non-obviousness, if present. *KSR*, 550 U.S. at 406.

“[W]hen a patent ‘simply arranges old elements with each performing the same function it had been known to perform’ and yields no more than one would expect from such an arrangement, the combination is obvious.” *Id.* at 417 (quoting *Sakraida v. Ag Pro, Inc.*, 425 U.S. 273, 282 (1976)). But in analyzing the obviousness of a combination of prior art elements, it can also be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *Id.* at 418.

“[I]n considering the disclosure of a reference, it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw

therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). Moreover, a precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *KSR*, 550 U.S. at 418. Rather, “any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.” *Id.* at 420. Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools International, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (quoting *Intelligent Bio – Sys., Inc. v. Illumina Cambridge, Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016)).

B. Level of Ordinary Skill in the Art

In our Institution Decision, we provisionally adopted Petitioner’s definition of a person of ordinary skill in the art (“POSA”) as having:

- 1) an advanced degree (such as an M.D. or the equivalent of a Ph.D. in a life or physical sciences, engineering, or mathematics discipline) . . .
- 2) an understanding of various sequencing and next generation sequencing methods, sample preparation methods, targeted enrichment methods, bioinformatics methods for interpreting sequencing data, and methods for identifying genetic variants in a sample
- 3) an understanding of clinical applications of sequencing technology to conditions associated with nucleic acid variation, such as cancer, non-invasive prenatal testing, and transplant rejection . . . [and]
- 4) an understanding of cancer biology as it relates to mutations associated with tumor diagnosis, progression, and treatment, as

well as awareness of the various types of mutations that can occur in and influence the course of a patient's cancer.

DI, 13–14 (citing, *e.g.*, Pet. 6; Ex. 1020 ¶¶ 25–38).

Patent Owner has not subsequently challenged the above definition. *See* Ex. 2031 ¶ 48 (Dr. Furneaux applying the above definition). For the reasons set forth in our Institution Decision, and because Petitioner's proposed level is commensurate with the level of skill reflected in the prior art of record, we apply the same definition here. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978).

C. Claim Construction

“Only when a claim is properly understood can a determination be made . . . whether the prior art anticipates and/or renders obvious the claimed invention.” *Amazon.com, Inc. v. Barnesandnoble.com, Inc.*, 239 F.3d 1343, 1351 (Fed. Cir. 2001). In this proceeding, we interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2020). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.* Furthermore, we need only construe claim language that is in controversy, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999); *see also Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (applying *Vivid Techs.* in the context of an AIA trial proceeding).

In this respect, Petitioner raises the issue of whether the preamble to claim 1 is limiting, proposes a construction for the term “polymorphism,” as used in claim step 1(a), and argues that the “biomedical report” of claim 2 is directed to printed matter not entitled to patentable weight. *See* Pet. 18–19, 29, 53–54. Because these terms are not in dispute, we decline to address them here. We address below the sole disputed claim term in this IPR—“capture probes.”

Claim step 1(b) involves “generating a second subset of nucleic acid molecules . . . by contacting said second nucleic acid sample with a plurality of capture probes, wherein said plurality of capture probes hybridize to one or more polymorphisms.” According to Patent Owner, “Foresight’s entire challenge to the ’033 patent hinges on a POSA conflating Forshew’s PCR primers with the claimed capture probes.” POR 66; *see also* Tr. 19:19 (“[a] central dispute is the construction of capture probes”).

Patent Owner argues that one of ordinary skill in the art would have understood “capture probes” to mean “a nucleic acid sequence that hybridizes to target DNA, either in solution or on a solid support, so that one can physically capture and isolate the target DNA.” *See generally* POR 54–66; Sur-reply 10–16. Applying its preferred construction, Patent Owner argues that “A POSA would not have considered Forshew’s PCR primers to be ‘capture probes’ because Forshew’s PCR primers do not hybridize to target DNA so that one can physically capture the target DNA.” POR 66.

For the purpose of institution, we found it sufficient to provisionally adopt Petitioner’s proposed definition of “capture probes” as encompassing at least, “PCR primer capture probes and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits).” *See* DI 18–24. For

the reasons set forth in our Institution Decision, and as further addressed below, we apply that definition here—albeit with a minor modification in wording. In particular, we credit Patent Owner’s argument that the phrase “PCR primer capture probe” is not a recognized term of art and its use in our construction “would improperly inject ambiguity into the claim.” POR 58–59 (citing, *e.g.*, Ex. 2018, 166:10–169:1, 171:16–175:2; *Liquid Dynamics Corp. v. Vaughan Co.*, 355 F.3d 1361, 1367 (Fed. Cir. 2004)). Although we recognize Petitioner’s use of “PCR primer capture probe” as a shorthand for its position that, as read in light of the Specification, PCR primers are a species of, and/or indistinguishable from, the genus of capture probes, Patent Owner’s point is well taken. Accordingly, we clarify the meaning of “capture probes,” as used in the instant Specification, to encompass PCR primers and solution-based hybridization capture probes (*e.g.*, biotinylated oligonucleotide baits). We do not view this modification as altering the substance of our provisional construction or the analysis in our Institution Decision.

In considering how one of ordinary skill in the art would have understood “capture probes” in the context of the ’033 patent, we look first to the Specification—“the single best guide to the meaning of a disputed term.” *In re Abbott Diabetes Care Inc.*, 696 F.3d 1142, 1149 (Fed. Cir. 2012) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1315 (Fed. Cir. 2005) (en banc)). The ’033 patent’s Specification provides no express definition of “capture probes,” but does, as Patent Owner points out, “describe[] several features of capture probes.” *See* POR, 56; *see also*, Tr. 20:1–21:19, 28:3–9 (“[O]ur specification does not provide a definition. It provides some description, but not a complete description”).

First, with respect to structure, the '033 patent teaches that, “capture probes may comprise *one or more nucleotides* . . . or 1000 or more nucleotides.” Ex. 1001, 31:35–43 (emphasis added); *id.* at 41:33–34 (“The capture probes . . . may comprise one or more nucleotides.”) In other words, capture probes within the scope of the '033 patent encompasses a nucleic acid range from a single nucleic acid to one of arguably infinite length.⁶

The Specification further discloses that a capture probe may *optionally* comprise one or more linkers and/or one or more labels. Ex. 1001, 31:4–12 (“The capture probe may further comprise one or more linkers. The capture probes may further comprise one or more labels. The one or more linkers may attach the one or more labels to the nucleic acid binding site.”). According to the Specification, the optional linker and/or label component of a capture probe encompasses a wide range of suitable materials:

Examples of linkers include, but are not limited to, hydrazone, disulfide, thioether, and peptide linkers. The linker may be a peptide linker. The peptide linker may comprise a proline residue. The peptide linker may comprise an arginine, phenylal[a]nine, threonine, glutamine, glutamate, or any combination thereof. The linker may be a heterobifunctional crosslinker.

Id. at 43:5–11; *see also id.* at 42:49–51 (“Suitable linkers comprise any chemical or biological compound capable of attaching to a label, primer, and/or capture probe disclosed herein.”)

With respect to “label” (interchangeable with “detectable label”), the Specification further teaches that these term’s “generally

⁶ Patent Owner agrees that the Specification encompasses capture probes having a length of one nucleotide. POR 62 (citing Ex. 2031 ¶ 69; Ex. 2027 (Gellert et al., “Helix Formation by Guanylic Acid,” 48 Proc. Nat. Acad. Sci. 2013–2018 (1962)).

refer() to any chemical moiety attached to a nucleotide.” *Id.* at 14:22–

29. More particularly,

labels include, but are not limited to, detectable labels, such as radioisotopes, fluorophores, chemiluminophores, chromophore, lumiphore, enzymes, colloidal particles, and fluorescent microparticles, quantum dots, as well as antigens, antibodies, haptens, avidin/streptavidin, biotin, haptens, enzymes cofactors/substrates, one or more members of a quenching system, a chromogens, haptens, a magnetic particles, materials exhibiting nonlinear optics, semiconductor nanocrystals, metal nanoparticles, enzymes, aptamers, and one or more members of a binding pair.

Id. at 23:23–36. The Specification further discloses that at least some linker molecules may also be labels. *See also id.* at 14:29–42 (further exemplifying labels as including “linker molecules” such as “biotin, avidin, streptavidin, HRP, protein A, protein G, antibodies or fragments thereof”). Illustrating the breadth of this term, the Specification “envision[s] that a change in mass may be considered a detectable label.” *Id.* at 14:42–47.

The Specification also refers to bead-bound nucleic acid molecules “associated with a nucleic acid molecule.” *Id.* at 14:18–19; 56–58.

Relatedly, the Specification teaches that “one or more subsets of nucleic acid molecules may be produced by contacting a nucleic acid sample with one or more beads, capture probes, labels, or a combination thereof.” *Id.* at 32:12–15. “The capture probes, primers, labels, and/or beads *may comprise one or more nucleotides*. The one or more nucleotides may comprise RNA, DNA, a mix of DNA and RNA residues or their modified analogs such as 2'-OMe, or 2'-fluoro (2'-F), locked nucleic acid (LNA), or abasic sites.” *Id.* at 41:32–37 (emphasis added). As such, the Specification instructs that the structure of capture probes (as well as primers, labels, and/or beads) encompasses a nucleic acid of any length, with or without the addition of other adducts—

which, themselves, may comprise a single nucleotide and/or a vast array of other entities.

Second, the Specification teaches that capture probes may also possess the functional attributes of binding or hybridization. Consistent with the plain language of step 1(a), “wherein said plurality of capture probes hybridize to a plurality of polymorphisms,” the Specification teaches that “[t]ypically, the capture probe comprises a nucleic acid binding site” and “may comprise a nucleic acid binding site that hybridizes to at least a portion of the one or more nucleic acid molecules or variant or derivative thereof in the sample or subset of nucleic acid molecules.” *Id.* at 31:7–8, 24–27. According to the Specification, “a capture probe hybridized nucleic acid refers [to] a capture probe associated with a nucleic acid molecule. The capture probe and the nucleic acid are in contact with each other.” *Id.* at 7:57–64, 14:48–59; *see also id.* at 22:65–23:9 (discussing genomic features/regions that may be hybridized by capture probes).

The Specification discloses that, “[c]apture probes may hybridize to *one or more* genomic regions” (*id.* at 22:65–23:1 (emphasis added)), but does not expressly define the size of a nucleic acid binding site, nor the size of a genomic region that needs to hybridize with the capture probe, nor binding conditions, nor any particular binding affinity for the capture probe.⁷ All that is required functionally, is that the capture probe “hybridizes to at

⁷ Given that the Specification teaches that a capture probe may be as short as one nucleotide, we might infer that the minimum size of its binding site and hybridized genomic region is also one nucleotide. *See* Ex. 1001, 31:35–43, 41:33–34; *see also* POR 62 (Patent Owner’s argument that “[t]he specification correctly includes capture probes having a length of one nucleotide because there are instances where a single nucleotide can function as a capture probe”).

least a portion of the one or more nucleic acid molecules or variant or derivative thereof in the sample or subset of nucleic acid molecules.” *Id.* at 31:25–27.

In short, all the Specification requires of capture probe function is that it hybridize to one or more nucleic acids. And as articulated at the institution stage, “[u]nder some (unspecified) conditions, and at some (unspecified) binding affinity, a single nucleic acid would be expected to hybridize with its complement and thereby meet the “capture probe” limitation as set out in the ’033 patent Specification.” DI, 44 (Jenks J., concurring).

In provisionally adopting Petitioner’s proposed definition in our Institution Decision, we first addressed Patent Owner’s contention that the ’033 patent “refers to capture probes and primers as separate and distinguishable from one another.” *See* DI 21–22; Prelim. Resp. 20–21 (citing, *e.g.*, Ex. 1001, 23:23–36; 32:56–60, 41:33–36). Considering the cited passages and, moreover, the entirety of the Specification, we were unable to discern any teaching as to *how* capture probes and primers might be “separate and distinguishable from one another,” as Patent Owner contends. *Id.* at 22. We noted, in particular, that

the Specification evinces no physical characteristics differentiating capture probes from primers. Both comprise from as few as “one or more nucleotides” to “1000 or more nucleotides” (Ex. 1001, 31:35–53, 40:28–54); both include nucleotides “compris[ing] RNA, DNA, a mix of DNA and RNA . . . modified analogs . . . locked nucleic acid (LNA), or abasic sites” (41:32–36); both may—or may not—comprise a detectable label (*id.* at 14:22–47, 23:23–36, 31:54–32:5, 40:29–35, 41:38–42:3); and both may—or may not—comprise a linker (*id.* at 31:8–9, 31:54–67, 40:29–31, 41:15–26, 42:49–43:11), wherein said linker can also be a detectable label (*id.* at 14:36–42).

DI 21–22.

With respect to function, we reasoned that because the Specification provides no physical difference between capture probes and primers “it would appear that the same polynucleotide could be employed as an amplification primer and a capture probe.” *Id.* at 22 (citing Prelim. Resp. 21). We likewise agreed with Petitioner that “none of the cited passages [by Patent Owner] describe ‘primers’ and ‘capture probes’ as mutually exclusive or precludes a similarly structured nucleic acid from performing each function.” *Id.* at 19; *see also* Reply 14 (citing POR 58, Ex. 1001, 23:23–26; Ex. 1225 ¶ 40).

We also noted Dr. Quackenbush’s testimony that dependent claim 10’s requirement that “capture probes further comprise barcodes,” typically indicated the use of PCR primers, but not hybrid capture baits. DI 19–20 (citing Ex. 1020 ¶¶ 102–103). And although Patent Owner responds that “barcodes can also be used on capture probes” (POR 63 (citing Ex. 2031 ¶ 70; Ex. 2028, 251)), this is not quite on point as Dr. Quackenbush and Mr. West subsequently confirmed that, while the use of barcoded PCR primers are well known, where bar codes are used in the context of biotinylated oligonucleotide bait systems, *the barcodes are not part of the capture probes*, rather, the barcodes are “part of the [captured DNA] molecule that’s being sequenced.” Ex. 1222, 167:20–170:4; Ex. 1225 ¶ 41.

As noted above, Patent Owner contends that its proffered definition of “capture probes” as meaning “*a nucleic acid sequence that hybridizes to target DNA, either in solution or on a solid support, so that one can physically capture and isolate the target DNA,*” excludes PCR primers, such as those employed by Forshew. *See generally* POR 55–67; Sur-reply 10–16.

In support of its construction, Patent Owner points first to the intrinsic record. Patent Owner argues that the Specification “treats capture probes and PCR primers as different types of nucleic acids,” that it separately describes the two terms, and, for some embodiments, “describe[s] the use of capture probes as a separate and distinct step from using PCR primers.” POR 58 (citing Ex. 1001, 31:4–32:5, 40:29–41:36, 23:23–26, 32:56–60, 64:29–43, 65:38–50, 66:53–67). Patent Owner made much the same argument prior to institution, but again, does not adequately describe any physical difference between the capture probes and PCR primers in its citations, nor persuade us that these entities, so defined in the Specification, are not functionally interchangeable. *See* DI 22 (concluding that “reading the Specification, it would appear that the same polynucleotide could be employed as an amplification primer and a capture probe in Patent Owner’s referenced embodiment(s)”).

Patent Owner further argues that during the prosecution history of the ’033 and ’394 patents, the Examiner⁸ agreed “that ‘pulldown probes’ are not PCR primers” and, in conducting prior art searches, the Examiner used the terms “capture” and “pull-down” synonymously, without including “primers” or “PCR” as search terms. POR 59 (citing Ex. 2004, 14, 17, 25; Ex. 2005, 19, 24, 31; Ex. 2024, 3–8, Ex. 2025, 3–25). But as previously noted, when the Examiner agreed that “pulldown probes” are not PCR primers, “the cited applications involved the claim limitation ‘pulldown probe,’ and claims reciting ‘capture probe’ were not before the Office.” DI 22–23 (“Although the cited interchange does not involve the claim term

⁸ Primary Examiner Kaijiang Zhang was the examiner of record in both the ’033 and ’394 patents.

“capture probe,” it shows the Examiner understood that ‘probes’ encompassed ‘amplification primers’.”). As such, Patent Owner’s argument remains unavailing.

As to Patent Owner’s assertion that the Examiner did not include “PCR” or “primers” in conducting prior art searches—but did apply “capture” and “pull-down” as synonyms—we note that the Examiner’s arguably synonymous search terms further encompassed “enrichment.” *See* Ex. 2024, 3–8 (“(enrich\$5 captur\$5 pull-down pulldown)”); Ex. 2025, 3–25 (same). As Patent Owner admits, the prior art recognized PCR as an enrichment technique. *See, e.g.*, POR 61 (citing Ex. 1006, 375), 63–64 (citing Ex. 1006, 378–80), 65–66 (“[A] POSA would understand that capture probes and PCR primers are different enrichment techniques”). As such, it would be reasonable to infer that the Examiner understood the search parameters to encompass PCR and PCR primers. Moreover, even if the Examiner had focused solely on the smaller genus of pull-down probes, this would still read on a claim more broadly reciting “capture probes,” properly construed. In any event, Patent Owner’s argument does not overcome the clear evidence in the Specification that the inventors chose to so broadly describe capture probes and PCR primers as to render the terms interchangeable. At a minimum, we find no persuasive evidence in the Specification or cited prosecution history that the scope of the term “capture probes” necessarily excludes PCR primers.

We also find persuasive Petitioner’s evidence that one of ordinary skill in the art would not have understood “capture probes” as synonymous with “pulldown probes.” *See* Reply 14–15. In this respect, Mr. West and Dr. Quackenbush agree that a “pulldown probe” refers to a biotinylated

oligonucleotide bait used in a “pulldown assay,” wherein the interaction between biotin and streptavidin attached to, e.g., a magnetic bead, “pulls down” or precipitates a molecule of interest. *See* Ex. 1222, 83:18–85:22; Ex. 1225 ¶ 39, 48. Patent Owner’s expert, however, testified that “capture probes” was a broader term, encompassing at least array-based hybrid capture, solution-based hybrid capture, guanine nucleotide—as taught by Gellert (Ex. 2027), Southern blot probes, and the zip-code PCR/ligase micro detection array probes taught by Gerry.⁹ *See* Ex. 2031 ¶¶ 65–70. Moreover, focusing on the intrinsic evidence, Petitioner notes:

The specification is explicit that [biotin and streptavidin] are considered “linkers” or “labels,” Ex1001, 14:22-46, 23:23-36, 31:7-12, which are optional and not necessary components of “capture probes.” Thus, molecules that include a nucleic acid binding site but lack any type of label – such as PCR primers – must fall within the scope of a “capture probe” as set forth in the Specification.

Reply 14–15 (further citing Ex. 1225 ¶¶ 38–39). We also note

Dr. Quackenbush’s reasoning that

[t]he specification is explicit that biotin and streptavidin are considered “linkers” or “labels,” ’394 patent, 14:22–46, 23:23–36, which are optional and not necessary components of “capture probes.” Thus, molecules that include a nucleic acid binding site but lack any type of additional chemical moiety (label or linker) such as biotin, must fall within the scope of a ‘capture probe’ as set forth in the Specification. This includes PCR primers, which are nucleic acids that hybridize to a target (i.e., contain a nucleic acid binding site), but which do not necessarily comprise linkers and/or labels.

Ex. 1225 ¶ 39.

⁹ Gerry et al., Universal DNA Microarray Method for Multiplex Detection of Low Abundance Point Mutations,” 292 *J. Mol. Biol.*, 251–262 (1999). (Ex. 2028).

Although the issued claims of the '033 patent recite “capture probes” rather than the “pull-down probes” limitation sought earlier in prosecution (*see* DI 29 (discussing Ex. 1078, 3; 2004, 17)), Patent Owner’s proposed construction requires that “capture probes” “*physically capture and isolate the target DNA*” (POR 56), and “exclusively . . . refer to hybridization capture” (Sur-reply 10). This construction is contrary to the Specification for the reasons set forth in paragraphs 43–51 of Dr. Quackenbush’s Second Declaration (Ex. 1225), which we adopt, and as further set forth below. In short, while the scope of “capture probe” set forth in the Specification clearly encompasses a biotinylated oligonucleotide bait and a magnetic bead system, which we agree may be used to physically capture and isolate a target DNA, Patent Owner chose not to describe or claim its invention so narrowly.¹⁰ To the contrary, the Specification discloses that the claimed “capture probes” may instead comprise a vast assortment of “linkers” and “labels” that suggest neither physical capture nor isolation of a nucleic acid, including nucleotides, radioisotopes, fluorophores, chemiluminophores, chromophores, electron donors/acceptors, dyes, spin labels, and even mere “change(s) in mass” detectable by surface plasmon resonance. *See, e.g.*, Ex. 1001, 14:22–47, 31:4–12, 41:32–42:37, 42:49–51 (“Suitable linkers comprise any chemical or biological compound capable of attaching to a label, primer, and/or capture probe disclosed herein.”). Consistent with the

¹⁰ Patent Owner cites TIP Systems for the proposition that “the claims of the patent need not encompass all disclosed embodiments. Our precedent is replete with examples of subject matter that is included in the specification, but is not claimed.” POR 60 (citing *TIP Systems, LLC v. Phillips & Brooks/Gladwin Inc.*, 529 F.3d 1364, 1373 (Fed. Cir. 2008)). But the present case represents the converse—where, the claims do not limit the scope of a term so broadly described in the Specification.

lack of any requirement for either physical capture or isolation, or indeed, “target DNA,” Petitioner correctly points out that the relevant limitations of claim element 1(b) require only “contacting” a nucleic acid sample with a capture probe, whereas the Specification teaches that any subsequent “separation” is a distinct “alternative[] or additional[]” step. *See* Reply 15–16 (citing Ex. 1001, 32:12–18, 26–33, 37:5–8; Ex. 1225 ¶¶ 42–47). As further discussed below, we read neither “capture probes,” nor the structure of the asserted claims to require a physical isolation. *See* Tr. 26:12–22 (Patent Owner’s counsel arguing that “physical isolation by pullout is implicitly required to do the sequencing of . . . limitations 1(b) and 1(c)”); *see generally id.* at 26:12–28:9.¹¹

Turning to extrinsic evidence, Patent Owner argues that one of ordinary skill in the art would recognize that PCR requires *two* primers (a primer pair) to amplify DNA, whereas using, e.g., biotinylated oligonucleotide baits and a magnetic bead system, “only *one* capture probe is required to physically pulldown a target DNA strand.” POR 60–61 (citing Ex. 1020 ¶¶ 72–76; Ex. 2018, 112:18–115:11; Ex. 2031 ¶ 66¹²). But while

¹¹ Patent Owner contends that “capture probes can be physically separated from [solid support] gels by cutting out the strip of nitrocellulose containing the captured nucleotides of interest and eluting the captured nucleotides.” POR 60 (citing Ex. 2031 ¶¶ 65; Ex. 2026). But even if “capture probes” and/or the “contacting” and “sequencing” limitations of claim elements 1(b) and 1(c) were to suggest a separation step, it was well known that PCR products could be similarly isolated, which further renders Patent Owner’s argument unavailing. *See* Reply 18–19 (citing Ex. 1225 ¶¶ 84–88; Ex. 2031 ¶ 65; Ex. 1223, 159:22–25, 160:22–161:3).

¹² Dr. Furneaux’s assertion that this indicates “physical differences between PCR primers and capture probes” is not persuasive insofar as nothing in the Specification prohibits a capture probe from comprising a primer or primer

the basic methodology of using PCR primers and biotinylated oligonucleotide baits are not in dispute, we find nothing in the Specification suggesting that a “capture probe” cannot encompass individual PCR primers and PCR primer pairs. *See, e.g.*, Ex. 1001, 32:12–15 (“[O]ne or more subsets of nucleic acid molecules may be produced by *contacting a nucleic acid sample with one or more* beads, *capture probes*, labels, or a combination thereof”) (emphasis added); *id.* at 41:32–33 (“The capture probes, primers, labels, and/or beads may comprise one or more nucleotides.”). Indeed, the Specification’s teaching that “[t]he methods disclosed herein may comprise 1 or more, 2 or more . . . capture probes or capture probe sets. The one or more capture probes or capture probe sets may be different, similar, identical, or a combination thereof,” indicates that “capture probe” should be read to encompass *both* individual PCR primers (where the one or more capture probes are identical), and PCR primer pairs (where the one or more capture probes are different). *See id.* at 31:13–23.

We are also not persuaded by Dr. Furneaux’s contention that “capture probes are uniquely different from PCR in that capture probes physically isolate native DNA.” *See* Ex. 2031 ¶ 74; *see also* Tr. 25:24–26:9 (Patent Owner’s counsel emphasizing that capture probes must isolate original target DNA rather than copies thereof). First, we credit Dr. Quackenbush’s testimony that this statement is not correct on its face because “[s]olution-phase hybridization capture baits are frequently used in experiments where the target DNA has been PCR amplified and modified with sequencing

pair. Moreover, claim steps 1(a) and 1(b) refer to a “plurality of capture probes,” which encompass one or more primers and/or primer pairs.

adaptors before capture.” *See* Ex. 1225 ¶ 50 (citing Ex. 1022, 91; Ex. 2031¹³ ¶ 62). Moreover, as stated by Petitioner, “both PCR primers and solution-phase baits are used to preserve sequence information and generate enough DNA for subsequent sequencing.” Reply 16–17 (citing Ex. 1225 ¶¶ 49–51); *see also* Ex. 1020 ¶¶ 66–78.

We further credit Petitioner’s evidence that persons of ordinary skill in the art understood PCR as a sequence capture technique. *See* Reply 17–18 (citing Ex. 1222, 95:15–18, 97:4–21, 170:21–24, Ex. 1030, 3 (PCR primer sets used “to capture the starting molecules present in the template”), Ex. 1216 (“The major differences with all the platforms are the way in which multiple loci are captured and amplified. Microfluidics, emulsion PCR and oligo-probes are the different ‘capture’ mechanisms”); Ex. 1217, 3–4 (“There are two basic approaches to sequence capture: PCR-based and hybridization-based, the latter of which can be done either in solution or on the surface of an oligonucleotide microarray.” (further referencing “more traditional PCR-based capture”); Ex. 1225 ¶¶ 54–57).

Relatedly, Patent Owner contends that one of ordinary skill in the art would understand that PCR primers have different functions and are categorized according to certain “core reaction principle[s].” POR 61–62.

According to Patent Owner,

the core reaction principle of capture probes involves hybridizing to “targeted regions of interest, either in solution or on a solid support, ***so that one can physically capture and isolate the sequences of interest.***” *Id.* In contrast, the core reaction principle of PCR primers is “***amplification***” of targeted

¹³ Gnirke *et al.*, “Solution hybrid selection with ultra-long oligonucleotides for massively parallel targeted sequencing,” 27(2) *Nature Biotechnology* 182–189 (2009). Ex. 1005.

regions by a “polymerase chain reaction (PCR)” that *copies* target regions. *Id.*; *see also id.* 376, Fig. 1. *Id.* (further citing Ex. 1020 ¶¶ 70–75; Ex. 2158, 2650; Ex. 2157, 320, Ex. 2157, 320; Ex. 2031 ¶¶ 67–68). Patent Owner similarly argues that one of ordinary skill in the art would select a particular enrichment procedure (e.g., PCR, on-array hybrid capture, in-solution hybrid capture) best suited to a particular research project or field of application. *See id.* at 63–66 (citing, e.g., Ex. 1006, 375, 378–80; Ex. 1006, 383, Ex. 1007, 114, Table 1; Ex. 2031 ¶¶ 71–74; Sur-reply 15–16. For the reasons set forth in paragraphs 52 and 53 of Dr. Quackenbush’s Second Declaration, which we adopt, we do not find Patent Owner’s argument persuasive. *See* Ex. 1225 ¶¶ 52–53. In short, that a person of ordinary skill in the art would evaluate the pros and cons of available enrichment techniques says little about whether PCR primers are within the scope of “capture probes” as described in the instant Specification.

For the reasons set forth above, and in light of all the evidence of record, we construe “capture probes,” as used in the instant Specification, to encompass PCR primers and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits).

D. Whether Forshew and Chan Qualify as Prior Art

Petitioner’s challenges are based on Forshew (Ground 1), Forshew in combination with Wagle (Ground 2), and Wagle in combination with Chan (Ground 3). Petitioner proffers Forshew and Chan, which presumptively constitute prior art under 35 U.S.C § 102(a) because they were published on May 30, 2012, and October of 2012, respectively, before the January 17, 2013, filing of the ’828 Provisional Application, upon which the ’033 patent

relies for priority. *See* Pet. 14–16 (citing Ex. 1020 ¶¶ 39, 94, 96); POR 5–6; Ex. 1001, code (60).

Patent Owner contends that the Petition fails because either Forshew or Chan are necessary for each asserted Ground, yet neither reference qualifies as prior art to the '033 patent. *See, e.g.*, POR 2, 5–54, 69, 77. To remove Forshew and Chan as prior art, Patent Owner undertakes the burden to show that the claimed invention was conceived before the effective date of the reference, followed by reasonably continuous diligence until constructive reduction to practice. POR 39–76 (citing, *e.g.*, *ATI Techs. ULC v. Iancu*, 920 F.3d 1362, 1369–70 (Fed. Cir. 2019)); Sur-reply 24. “Demonstration of such priority requires documentary support, from which factual findings and inferences are drawn, in application of the rules and law of conception, reduction to practice, and diligence.” *ATI Techs* at 1369 (quoting *In re Steed*, 802 F.3d 1311, 1316 (Fed. Cir. 2015)). Conception, for example, “must be proved by corroborating evidence which shows that the inventor disclosed to others his completed thought expressed in such clear terms as to enable those skilled in the art to make the invention.” *Coleman v. Dines*, 754 F.2d 353, 359 (Fed. Cir. 1985) (internal quotation marks omitted). “The conception analysis necessarily turns on the inventor’s ability to describe [the] invention with particularity.” *Burroughs Wellcome Co. v. Barr Labs., Inc.*, 40 F.3d 1223, 1228 (Fed. Cir. 1994) (explaining that “[a]n idea is definite and permanent when the inventor has a specific, settled idea, a particular solution to the problem at hand, not just a general goal or research plan he hopes to pursue”). Until the inventor can do so, the inventor “cannot prove possession of the complete mental picture of the invention.” *Id.*

Patent Owner contends that Forshew and Chan do not qualify as prior art because “the inventors of the ’033 patent conceived of the claimed invention by no later than May 8, 2012 and diligently worked towards its reduction to practice via filing the [’828] [P]rovisional [A]pplication on January 17, 2013,” which, if established, would mean the claimed invention has a priority date of May 8, 2012—prior to the publication of either Forshew or Chan. POR 6.

With respect to conception, Patent Owner asserts that the claimed invention was developed through Personalis’ “Accuracy Program.” POR 7. Patent Owner relies on Mr. West’s testimony that, “[m]y co-inventors and I met many times between November 2011 and May 8, 2012 to discuss ideas for the Accuracy Program.” *See* POR 10–12; Ex. 2032 ¶ 57. In support, Mr. West points to an email (with attachments) dated January 6, 2012, as indicating that his team was working on “‘augment[ing] NGS [next-generation sequencing] with other technologies to determine key variants,’ and ‘conduct[ing] follow-up genetic & non-genetic testing to validate results.’” Ex. 2032 ¶¶ 57–59 (citing Ex. 2046, 1 (cover email), 3 (“slide 2”), 10 (“slide 9”), 14 (“slide 13”)). Further referencing Mr. West’s testimony, Patent Owner argues that “[o]n January 13, 2012, Mr. West planned to supplement whole genome sequencing using hybrid capture ‘oligo-based pullouts.’” POR 11 (citing Ex. 2032 ¶¶ 53–56; Ex. 2109, 2 (“Targeted re-sequencing using array or oligo-based pullout”), 3 (“Exon re-sequencing by pullout . . . Additional re-sequencing by pullout, to capture regions outside the exons”); Ex. 2102, 2).

Patent Owner argues that “[o]n April 2, 2012, Mr. West described Personalis’ laboratory plans.” *Id.* (citing Ex. 2036; Ex. 2032 ¶ 61).

“Inventors met and exchanged ideas for the Accuracy Program on May 3 and 4, 2012.” *Id.* (citing Ex. 2037; Ex. 2038; Ex. 2039; Ex. 2040; Ex. 2032 ¶¶ 62–65). And “[o]n May 8, 2012, Mr. West presented his ‘CEO Overview’ slides to the Personalis Board of Directors.” *Id.* at 11–12 (citing, e.g., Ex. 2033 ¶¶ 20–26; Ex. 2112). Patent Owner relies on Mr. West’s “CEO Overview,” and in particular Slide 5 (reproduced below), as evidence that the inventors had conceived of the claimed invention as of May 8, 2012. *See generally id.* at 12–39; Sur-reply 4–9.

Overview of Personalis Accuracy Program

Characterize errors	Informatic Solutions	Lab, Content Solutions
<u>Analysis</u> Rates by variant type & Q-score Underlying mechanisms Medical interpretation impact Regulatory impact	<u>Advances in 3rd party alg's</u> Aligner for > 200bp reads New versions, current alg's Other alg's (e.g. lobSTR)	<u>Combine technologies</u> WGS + exome Longer reads and/or inserts Multi-vendor Arrays (genotypes & SV's) Electrophoretic sizing Sanger validation
<u>Methods</u> Comparison with literature Studies on limited regions Whole genome analysis: Single genome metrics Same person twice Family inheritance (MIE) Platform comparison Arrays ILMN / CGI / Ion T... Sanger capillary	<u>Advances in our alg's:</u> SV's : Zygosity, accuracy Combine SV's & SNP's Broader use of Breakseq	<u>Content-based semi-custom</u> Array (genotypes & SV's) Pullout
<u>Prioritize solutions</u> Bio-informatic approaches Laboratory approaches Content approaches	<u>New Personalis alg's</u> Alignment degeneracy Genome-aware alignment VNTR-specific	<u>Custom sample preps</u> Problematic motifs Degenerate regions Phasing
	<u>Advances in reference</u> Ethnic major-allele ref's Iterative Admixed InDel's / SV's	<u>Content</u> Gold standard genomes Known healthy genomes
	<u>Flag or Filter Out</u> For problems not yet solved	

According to Patent Owner, Slide 5 “concisely describes the Accuracy Program.” *Id.* at 13 (citing Ex. 2112, 5).

Petitioner, however, challenges the credibility of Personalis’ invention story. *See generally* Reply 1–2; Tr. 10:19–13:18. As an initial matter, Petitioner notes that named co-inventor “ Dr. Bartha started work at Personalis on May 1, 2012—just a few days before its *latest* proffered bound for conception.” Reply 3. According to Petitioner, it is “highly implausible

that Dr. Bartha participated in conception within just a few days of starting at Personalis,” particularly given that “Mr. West’s May 8 Board of Director[’]s presentation, which is the basis for Personalis’s ‘no later than’ date, was drafted prior to May 8, . . . and Dr. Bartha is not listed as a participant at the May 3 and May 4 meetings Mr. West discusses.” Reply 3–4 (citing Ex. 1222, 51:18–24; Ex. 2038–2040; Ex. 2032 ¶¶ 63–65).

In response, Patent Owner points to Mr. West’s testimony that upon joining Personalis, Dr. Bartha “was immediately tasked with working on the Accuracy Program under Mark Pratt.” Sur-reply 4 (citing Ex. 2032 ¶ 34). But other than at this high level of generality, Patent Owner provides no evidence of what tasks Dr. Bartha undertook or how Dr. Bartha allegedly contributed to any claim of the challenged patent.¹⁴

Also pertinent to credibility, Patent Owner’s reliance on the May 8, 2012 “CEO Overview” is somewhat at odds with Mr. West’s implication that conception was at least substantially complete more than a month before Dr. Bartha’s start date. In particular, Mr. West testified that

all the pieces that came together to form the basis for the ’033 and ’394 patent, probably the last of those elements to come together was from a paper that was -- I believe it’s an exhibit here – that was published by Mike Snyder’s group at Stanford.

¹⁴ With respect to Petitioner’s related argument that co-inventor Luo was not hired until June 18, 2012, Patent Owner argues that Ms. Luo is only an inventor with respect to the “variable length tandem repeat” aspect of dependent claim 4. Reply 2–3; Sur-reply 4–5; Tr. 10:24–11:7, 12:3–11, 29:2–30:4. Although we recognize Petitioner’s argument that Ms. Luo only contributed to reduction to practice of claim 4, not conception, because Patent Owner’s antedation proofs address only the May 8, 2012 conception date for independent claim 1, for the purpose of this Opinion, it is sufficient that we apply our conclusion regarding the lack of conception of claim 1 to claim 4 and the other dependent claims.

The publication came out in March of 2012, and our group had seen a preview copy of it at least a month earlier. We worked very closely with Mike since he was a cofounder of the company.

Ex. 1222, 48:19–49:3. And although Mr. West later stated that “the last of the elements that, then, came together in the material that we presented to our board on May 8th,” the record does not explain what “last of the elements” pertinent to conception were added in the CEO Overview, nor what contribution Dr. Bartha made to the claimed invention. *See id.* at 49:12–14.

Relying on the testimony of Dr. Quackenbush, Petitioner next contends that the Accuracy Program cited as the basis for Personalis’ conception, was not a program to develop a minimum residual disease test such as the NeXT Personal MRD assay Patent Owner asserts is covered by the claims, but “a program to improve sequencing accuracy.” Reply 4–5 (citing, *e.g.*, Ex. 1225 ¶¶ 116–125); POR 80. “Hence,” Petitioner contends, “Personalis’s first product – prototypes of which the patent expressly states were the source of its data – was its Exome Plus/Ace Exome product, released in February 2013,” whereas “Personalis’s NeXT Personal MRD diagnostic test was not launched until 2021,” nine years after Mr. West’s 2012 presentation. Reply 4–5 (citing Ex 2032 ¶38; Ex. 1001, 74:60–64); Ex. 1225 ¶ 119. While not dispositive, this testimony supports Petitioner’s contention that work related to the Accuracy Program was not directed to conception of the invention claimed in the ’033 patent.

Further supporting Petitioner’s contention that Patent Owner’s documents do not evidence conception of the claimed invention, Dr. Quackenbush credibly testifies that one of ordinary skill in the art would understand references in Exhibit 2046 to, for example, “characteriz[ing]

errors” and performing “whole genome sequencing” on the “same person twice,” and “Pullout” in Slide 5, as well as “Repeatability,” and “Same person twice (or more?), trying to keep everything else constant,” as directed to improving sequencing accuracy rather than to a program to develop a diagnostic assay relating to the inventions described in the challenged method claims. *See* Ex. 1225 ¶¶ 119–125); Reply 4–5.

In contrast, Dr. Furneaux offers no independent opinion on how one of ordinary skill in the art would have understood Patent Owner’s prior conception evidence, or whether either Forshew or Chan is prior art. *See* Ex. 2031 ¶¶ 190, 203, 220. On balance, Dr. Quackenbush’s testimony weighs against a finding that Patent Owner’s evidence demonstrates a conception of the invention claimed.

Noting that Patent Owner’s conception proofs “rel[y] exclusively on Ex2112, slide 5 for the ‘plurality of capture probes’ limitation of step 1(a),” Petitioner contends that its “bare reference to ‘pullout’” suggests at most, “that by May 2012, Mr. West contemplated that a particular type of capture probe would be among many other ‘lab solutions’ for addressing accuracy.” Reply 7–8; *see* Ex. 1225 ¶¶ 123–126. In this respect we note Mr. West’s testimony that he understood “pullout” in Slide 5, as a reference to Agilent SureSelect biotinylated probe system. *See* Ex. 1222, 83:8–85:22, 88:7–16. We, nevertheless, find persuasive Dr. Quackenbush’s testimony that “a POSA reviewing Slide 5 in light of the overall record on conception would have insufficient information to determine how, specifically, Personalis intended to use the ‘Pullout,’ other than that their purpose was to provide a laboratory solution to improve the accuracy of 2012 sequencing technology in some way.” Ex. 1225 ¶ 125.

Considering the evidence of record, we credit Dr. Quackenbush's testimony that a person of ordinary skill in the art reading Personalis' conception exhibits, including Exhibit 2112, "would have found, at most, an incipient research project for improving the accuracy of sequencing technology that would involve the use of an undefined combination of various conventional techniques." Ex. 1225 ¶ 128. In addition to Dr. Quackenbush's testimony as to claim steps involving capture probes (elements 1(a)–1(c)), we further find persuasive his testimony that Patent Owner provides no corroboration of the conception of steps 1(d)–1(f), which require assaying samples obtained from the same individual obtained *at different time points*. *See id.* ¶¶ 120–121, 126–127; Reply 9–10; Sur-Reply 6–8 (citing, *e.g.*, Ex. 2032 ¶¶ 91–134; Ex. 2109; Ex. 2112; Ex. 2036). In sum, we find that Personalis' exhibits and related testimony do not evidence sufficient conception of the invention set forth in the challenged claims. *See, e.g.*, Ex. 1225 ¶¶ 120–128 (Dr. Quackenbush's testimony about missing aspects of the claims in Patent Owner's proofs, such that the "completed thought" is not shown).¹⁵

Considering the above, we find that Patent Owner has not satisfied the burden of establishing that its claimed invention is entitled to a priority date that precedes either of the respective publication dates of Forsheew and Chan. Accordingly, Forsheew and Chan qualify as prior art to the '033 patent.

¹⁵ For the purpose of this Decision, we need not address Petitioner's further contention that the filing of the '828 Provisional Application does not qualify as a constructive reduction to practice of the challenged claims. *See* Reply 13; Sur-reply 9.

E. Ground 1: Anticipation by Forshew

As to Ground 1, Petitioner challenges claims 1–7 and 10–23 as anticipated by Forshew. Pet. 11, 24–46. Petitioner provides an element-by-element comparison of the claims to the cited art. *Id.* at 29–46; Reply 13–20; Resp. 1–3; Ex. 1020 ¶¶ 107–172; Ex. 1225 ¶¶ 101–103.

Patent Owner opposes. *See* POR 54–68; Sur-reply 10–16; Ex. 2007 ¶¶ 12–35; Ex. 2031 ¶¶ 189–200. We address with particularity only those elements raised by Patent Owner, finding the remaining elements supported by Petitioner’s un rebutted arguments and evidence. We begin with an overview of the asserted reference.

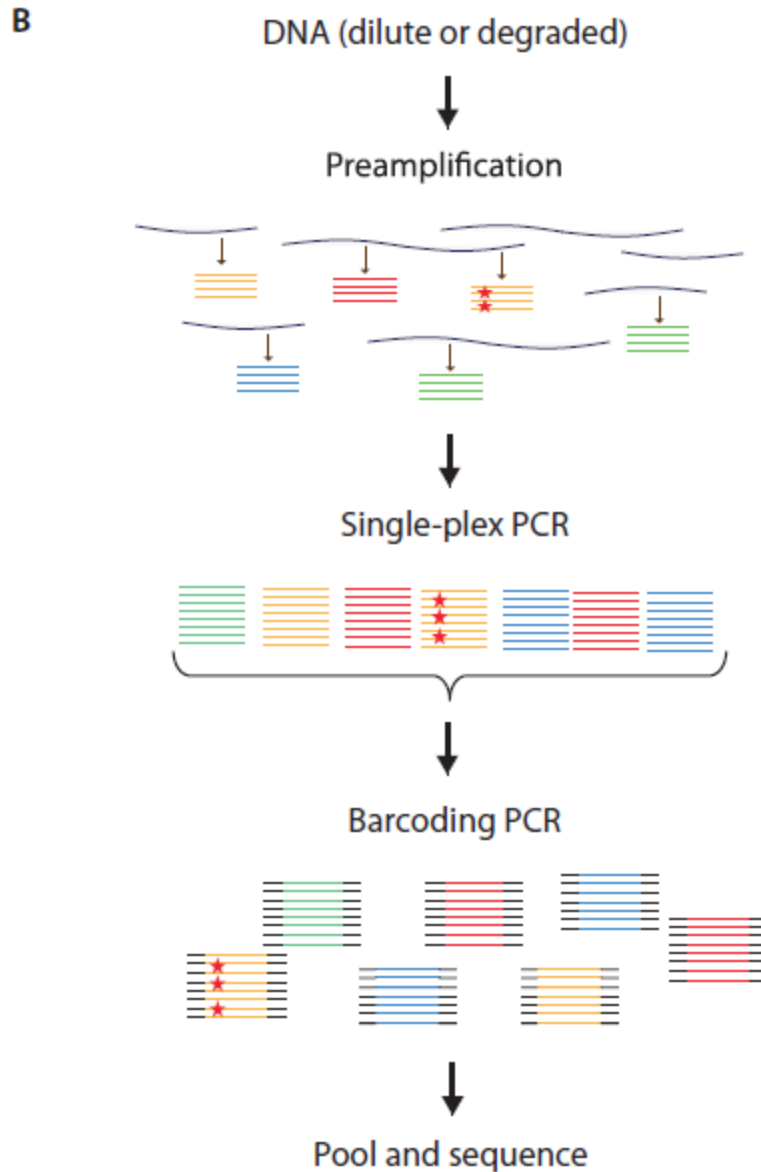
1. Overview of Forshew (Exhibits 1030, 1032)

Forshew employs tagged-amplicon deep sequencing (TAm-Seq) for the detection and monitoring of cancer-associated alleles circulating in patient plasma samples. Ex. 1020, 1. According to Forshew, the TAm-Seq approach “allows amplification and deep sequencing of genomic regions spanning thousands of bases from as little as individual copies of fragmented DNA,” which “allowed us to monitor changes in tumor burden by sampling only patient plasma over time.” *Id.* at 1–2; *see also id.* at Figs. 4B–D (graphics illustrating monitoring frequency of mutant DNA in cancer patients over time).

To amplify and sequence fragmented cell-free DNA in patient plasma samples, Forshew used the Primer3 program¹⁶ to “design[] primers to

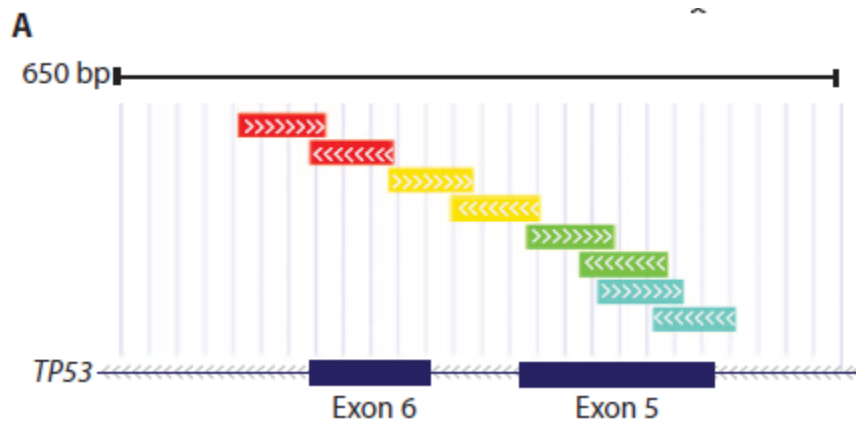
¹⁶ It is undisputed that Primer3 “was and continues to be a widely-used open source program that can pick PCR primer sequences using criteria such as oligonucleotide melting temperature, size, GC content, primer-dimer possibilities, PCR product size, and amplifying the wrong sequence.”

generate amplicons that tile regions of interest in short segments of about 150 to 200 bases” for cancer-related genes including *TP53*, *EGFR*, *BRAF*, and *KRAS*. *Id.* at 2, Ex. 1032, 2. Forshev illustrates the generation of amplicons for a targeted genomic region in Figure 1B, reproduced below.



Ex. 1020 ¶ 109 (citing <http://primer3.org>; Exhibit 1034 (Untergasser et al., “Primer3–new capabilities and interfaces,” 40(15) NAR e115 (2012))).

Ex. 1030, Fig. 1B. Figure 1B shows a multi-step PCR amplification process including the addition of barcodes and sequencing adaptors, wherein amplified, barcoded PCR products are pooled and sequenced. *See id.* at legend. Red stars in some amplicons indicate the presence of mutations. *Id.* Amplicon sequences are compared and aligned based on partial overlaps, as shown in Figure 1A, reproduced below.



Ex. 1030, Fig. 1A. Figure 1A shows the location of eight overlapping amplicon "tiles" spanning a 650 base pair region of the TP53 gene, which includes exons 5 and 6. *See id.* at legend. Arrowheads within tiles indicate direction of sequencing read. *Id.*

Forsheew discloses that, for the TAm-Seq studies,

We designed a set of 48 primer pairs to amplify 5995 bases of genomic sequence covering coding regions (exons and exon junctions) of TP53 and PTEN, and selected regions in EGFR, BRAF, KRAS, and PIK3CA (table S1) by overlapping short amplicons (Fig. 1A). The sequenced regions cover mutations that account for 38% of all point mutations in the COSMIC database (v55).¹⁷

¹⁷ It is undisputed that, "COSMIC (Catalogue of Somatic Mutations in Cancer), was and continues to be a leading and comprehensive expert-curated database of somatic mutations in human cancers." Ex. 1020 ¶ 111 (citing <https://cancer.sanger.ac.uk/cosmic>; Ex. 1035 (Forbes et al,

Id. at 3. According to Forshew, the resulting data was used to “identif[y] mutations throughout the tumor suppressor gene TP53 in circulating DNA from 46 plasma samples of advanced ovarian cancer patients;” “noninvasively identify the origin of metastatic relapse in a patient with multiple primary tumors;” identify “an *EGFR* mutation not found in an initial ovarian biopsy;” and “monitor tumor dynamics, and track[] 10 concomitant mutations in plasma of a metastatic breast cancer patient over 16 months.” *Id.* at Abstract.

2. “capture probes”

Claim step 1(a) involves “producing, with the aid of a computer processor, a plurality of capture probes, wherein said plurality of capture probes hybridize to a plurality of polymorphisms.” Claim step 1(b) recites “contacting said nucleic acid sample with said plurality of capture probes produced in (a).” Patent Owner contends that Forshew fails to disclose the “capture probes” or “plurality of capture probes” limitations of claim 1 under its preferred construction. POR 66–67; Sur-reply 10–16.

In Section II.C. above, we construe “capture probes,” as used in the instant Specification, to encompass PCR primers and solution-based hybridization capture probes (e.g., biotinylated oligonucleotide baits). Accordingly, in disclosing the use of PCR primers, Forshew discloses “capture probes,” as would have been understood by one of ordinary skill reading the ’033 Specification. *See* Pet. 29–33 (citing Ex. 1020 ¶¶ 117–122; Ex. 1030, 2–3). Forshew further satisfies the “plurality of capture probes”

“COSMIC: mining complete cancer genomes in the Catalogue of Somatic Mutations in Cancer,” 39 NAR D945-950 (2011)).

limitation in disclosing the use of PCR primers for the detection and monitoring of cancer-associated alleles. *See, e.g., id.*; Ex. 1020, 3 (“We designed 48 primer pairs to amplify 5995 bases of genomic sequence covering coding regions (exons and exon junctions) of TP53 and PTEN, and selected regions in EGFR, BRAF, KRAS, and PIK3CA (table S1) by overlapping short amplicons (Fig. 1A).”).

For the reasons set forth above, Petitioner has sufficiently shown that Forshew discloses “capture probes” and “a plurality of capture probes,” as would have been understood by one of ordinary skill reading the ’033 Specification.

3. “conducting a sequencing assay”

Claim step 1(c) requires “conducting a sequencing assay on a subset of nucleic acid molecules generated from said nucleic acid sample in step (b) to yield a result comprising a nucleic acid sequence, thereby analyzing said nucleic acid sample.” With respect to this step Petitioner and Dr. Quackenbush point to, for example, Forshew’s Figure 1B as illustrating Forshew’s teaching to “‘pool and sequence’ the amplicons generated from the nucleic acid sample in [claim step 1b]. Pet. 33–35; Ex. 2020 ¶ 127.

According to Patent Owner, “Forshew does not anticipate this step because its sequencing assay is conducted on molecules that result from PCR amplification—not on molecules captured by contacting the capture probes with the nucleic acid sample.” POR 67–68; Sur-reply 16 (similar).

We do not find Patent Owner’s argument availing for substantially the reasons set forth above with respect to claim steps 1(a) and 1(b). Because “capture probes,” as used in the ’033 patent encompasses Forshew’s PCR primers, we agree with Petitioner that the “pool and sequence” step

illustrated in Forshew's Figure 1B satisfies the "conducting a sequencing assay on a subset of nucleic acid molecules generated from said nucleic acid sample in step (b) to yield a result comprising a nucleic acid sequence, thereby analyzing said nucleic acid sample," recited in claim step 1(c).

4. "repeating steps (b)–(c)"

Claim step 1(d) entails "repeating steps (b)–(c) on a subsequently obtained biological sample from said individual." For this limitation, Petitioner points to, for example, Forshew's Figure 4B, which reports "[m]onitoring the frequency of mutant DNA in plasma of an ovarian cancer patient (Patient 46) over time using T[A]m-Seq and digital PCR." Pet. 36 (citing Ex. 1030, Figs. 4B–C; Ex. 2020 ¶ 131).

Patent Owner contends that "Forshew does not disclose this step because it never conducts steps (b) and (c) to begin with, let alone repeats them." POR 68. We do not find Patent Owner's argument persuasive. For the reasons set forth above, Petitioner has shown sufficiently that Forshew discloses claim steps 1(b) and 1(c). And the present record further indicates that Forshew discloses examples wherein those steps are repeated on a subsequently obtained biological sample from an individual as required by claim step 1(d). *See* Ex. 1030, 2 (stating that the disclosed method "allowed us to monitor changes in tumor burden by sampling only patient plasma over time," Figs. 4B–D (graphics illustrating monitoring frequency of mutant DNA in cancer patients over time); Ex. 2020 ¶¶ 130–131.

5. Conclusion as to Ground 1, Claims 1–7 and 10–23

Considering the record before us, and in light of our construction of "capture probes," we find that Petitioner has shown persuasively that claims 1–7 and 10–23 are anticipated by Forshew.

F. Ground 2: Obviousness in View of Forshew and Wagle

As to Ground 2, Petitioner further challenges claims 1–23 as unpatentable for obviousness in view of Forshew, as discussed above, and further in view of Wagle. *See* Pet. 16, 46–51; Reply 22–27; Ex. 1020 ¶¶ 173–190; Ex. 1225 ¶¶ 104–107. Patent Owner opposes. *See* POR 68–76, 79–83; Sur-reply 16–23; Ex. 2031 ¶¶ 201–218. We address with particularity only those elements raised by Patent Owner, finding the remaining elements supported by Petitioner’s unrebutted arguments and evidence. As Forshew is discussed above, we begin with an overview of Wagle.

1. Overview of Wagle (Exhibit 1033)

Wagle discloses a “a targeted, massively parallel sequencing approach to detect tumor genomic alterations in formalin-fixed, paraffin embedded (FFPE) tumor samples.” Ex. 1033, 83. Wagle explains that, “hybrid selection-based methods that enrich for coding sequences prior to sequencing (“exon capture”) are routinely being implemented in discovery-oriented settings. Here, we describe an adaptation of exon capture and massively parallel sequencing for robust detection of somatic genomic alterations in FFPE samples.” *Id.* at 84 (references omitted).

According to Wagle:

Massively parallel sequencing libraries (Illumina) that contain barcoded universal primers (9) were generated with the use of genomic DNA from formalin-fixed, paraffin-embedded tumor material. After preamplification and DNA quantification, equimolar pools were generated consisting of 12 barcoded tumor DNAs. These DNA pools were subjected to solution-phase hybrid capture with biotinylated RNA baits targeting all exons from 137 actionable cancer genes. Each hybrid capture

reaction was sequenced in a single paired-end lane of an Illumina flow cell.

Id. at 91.

In particular, Wagle “generated a list of 137 ‘druggable’ or potentially actionable genes known to undergo somatic genomic alterations in cancer.”

Id. at 84. Wagle then used a commercially-available computer program to design “7,021 unique biotinylated RNA baits corresponding to these genomic regions,” such that “[t]arget loci were covered with a tiling density of $\times 2$.” *Id.* at 84, 91 (“The Agilent SureSelect E-array program was used to design 7,021 unique RNA baits corresponding to the coding sequence of the 137 [cancer-related] genes described above, as well as to the 79 pharmacogenomic polymorphisms and to 24 SNPs for fingerprinting.”).

Wagle further discloses that the baits were replicated 8 times on a 55,000-bait library array. *Id.* at 91.

Using this approach, Wagle detected single nucleotide variants, insertion/deletions and copy number alterations in a collection of cancer cell lines. *E.g.*, Ex. 1033 at 84 (“In total, 102 single-nucleotide variants and 6 indels (excluding known germline polymorphisms) were detected in coding sequences across the 10 cell lines, including all 21 single-nucleotide variants and 3 of 4 indels reported for these lines in the Catalogue of Somatic Mutations in Cancer database (COSMIC)”; 86 (“Comparison of overall copy number values derived by sequencing to those obtained from high density single-nucleotide polymorphism (SNP) array data (Affymetrix SNP 6.0 platform) demonstrated a robust correlation at the gene level”).

2. Overview of Petitioner’s Arguments

With respect to claims 1–7 and 10–23, Petitioner relies on Wagle as teaching capture probes, but only to the extent Forshew’s PCR primers were

not within the scope of “capture probes” as understood in the context of the ’033 patent. Pet. 46–49. Further analysis of this position is unnecessary in view of our construction of this term and our finding that claims 1–7 and 10–23 are anticipated by Forshew. *See* Sections II.C and E, above; *Realtime Data, LLC. v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (“It is well settled that a disclosure that anticipates under § 102 also renders the claim invalid under § 103, for anticipation is the epitome of obviousness.”) (internal quotation marks and citations omitted).

Petitioner also relies on Wagle as disclosing the “hybridization array” and “plurality of capture probes . . . conjugated to beads” recited in claims 8 and 9, respectively. Pet. 49–51. According to Petitioner, Wagle “disclos[es] the use of RNA bait hybridization probes in a similar context to Forshew’s method.” *See id.* at 47. As articulated by Dr. Quackenbush, Petitioner contends that one of ordinary skill in the art “would have been motivated to combine Forshew and Wagle by using Wagle’s solution-phase hybridization RNA baits as capture probes, either in place of, or in addition to, Forshew’s PCR primer capture probes,” because (1) “a POSA would have considered these amplification techniques to be relatively interchangeable;” (2) in light of Wagle’s “suggest[ion] that in certain circumstances, PCR-capture based techniques may be less optimal (although not ineffective);” and (3) in view of Wagle’s proposal to use “its exon capture approach as a related technique suitable for the capture portion of the general workflow.” *See* Ex. 1020 ¶ 182 (footnote omitted).

In addressing reasonable expectation of success, Dr. Quackenbush further testifies that Forshew’s PCR and Wagle’s RNA bait-based hybridization “techniques are not mutually exclusive, as a POSA could have

been motivated to use hybridization followed by PCR, or vice-versa, in certain applications, for example, in some scenarios in order to further enrich a target sequence to[]further improve and refine results as compared to performing either step alone.” *Id.* ¶ 184. In support of this position, Dr. Quackenbush reproduces Figure 2 of Mamanova¹⁸ to illustrate that solution-based hybrid capture and PCR were two well-known types of target enrichment techniques that could be applied at the choice of the POSA, and in some scenarios the techniques were known to have overlapping applicability. Ex. 1020 ¶ 184 (citing Ex. 1007).

¹⁸Mamanova *et al.*, “Target-enrichment strategies for next-generation sequencing,” 7(2) *Nature* 111–118 (2010). Ex. 1007.

Mamanova's Figure 2 is reproduced below.

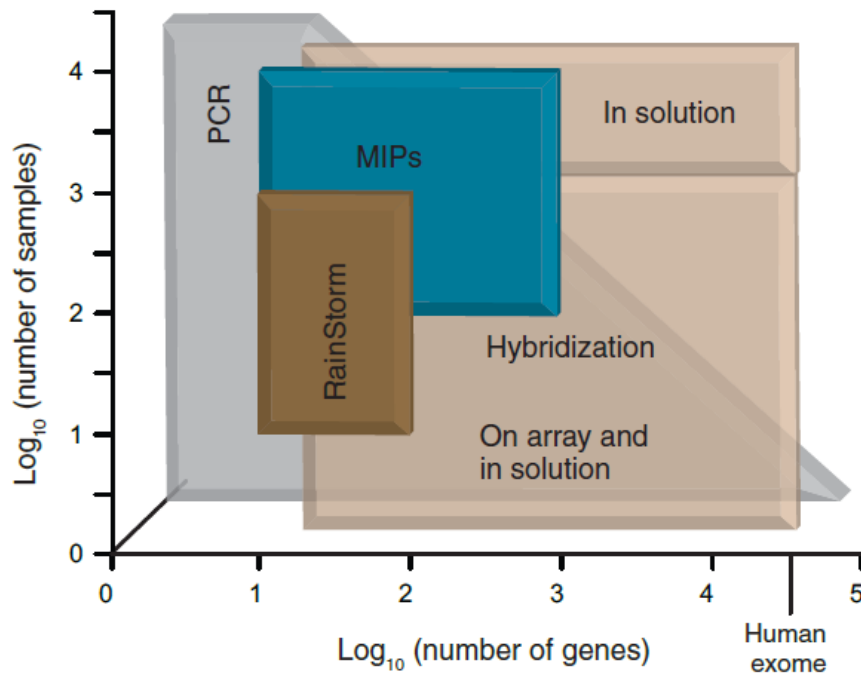


Figure 2 | Suitability of different target-enrichment strategies to different combinations of target size and sample number. Suitability was estimated from the perspective of the feasibility with which each method could be applied to the various combinations of target size and sample number, rather than the cost.

Ex. 1007, Fig. 2. Figure 2 of Mamanova is a graph, plotting the number of genes (x-axis) against the number of samples (y-axis), with both variables in Log₁₀ scale, and provides an estimate of “suitability” for different target-enrichment strategies “from the perspective of the feasibility with which each method could be applied to the various combinations of target size and sample number, rather than cost.” *Id.* at legend. Mamanova's Figure 2 shows, for example, the suitability of PCR (gray, triangular shading) and on-array and in-solution hybridization (beige, square shading) as a function of number of samples and number of genes to be analyzed.

Dr. Quackenbush further testifies that one of ordinary skill in the art would have reasonably expected to succeed in using Wagle's RNA bait-based probes in place of Forshew's PCR primer capture probes in the capture step. Both techniques essentially involve enrichment of targeted sequences using hybridization of probes to targets. Wagle's approach enriches by isolating the targets and removing them from non-targeted nucleic acids in the sample. Forshew's approach captures the targets through base-pairing hybridization and copies the targets and thereby enriches them over non-targeted nucleic acids.

Id. ¶ 183.

3. Motivation to Combine

In response, Patent Owner contends that one of ordinary skill in the art would have had no reason to combine the PCR-based method of Forshew with Wagle's use of hybridization-based biotinylated capture baits. *See generally* POR 68–76, 79–83; Sur-reply 16–23; Ex. 2031 ¶¶ 201–162. We address those arguments below.

a) Required Amounts of DNA

Patent Owner first asserts that the amount of DNA used in Forshew's PCR-based system is incompatible with the use of hybridization-based biotinylated baits taught by Wagle. POR 69–72; Sur-reply 20–21. Patent Owner contends that Forshew's PCR-based assay is designed to operate on “minute” (0.9–19.7 nanograms) amounts of DNA, equivalent to that present in from 30 to 150 microliters of blood plasma. *Id.* at 69–70 (citing Ex. 2031 ¶¶ 204–207; Ex. 1030, 4–6, Tabs. 1, 2). According to Patent Owner “the amount of DNA used in Forshew's T[A]m-Seq assay is so tiny that *three rounds* of PCR amplification are required to generate sufficient amounts of DNA”—15 cycles of “preamplification,” 35 cycles of “single-plex PCR,” and 15 cycles of “barcoding PCR.” *Id.* at 70 (citing Ex. 1030, 2–3, Fig. 1(B);

Ex. 1032, 2–3; Ex. 2031 ¶ 207). Patent Owner contrasts the less than 20 nanogram amounts of plasma DNA used by Forshew with the 3,600 ng of tumor-derived DNA used in Wagel’s hybrid selection step as evidence that one of ordinary skill in the art “would not have been motivated to replace Forshew’s PCR-based amplification with [hybridization-based biotinylated baits] with a reasonable expectation of success because [hybridization-based biotinylated baits] do not amplify DNA.” *Id.* at 70.

We do not find Patent Owner’s argument persuasive for the reasons set forth on pages 21–22 of the Reply; *see also id.* at 23–24 (arguing that obviousness does not require a physical combination of the two systems). First, Forshew worked with small quantities of DNA because a goal of the PCR-based TAm-Seq method was to “sequence large genomic regions from even single copies of ctDNA,” and contains no prohibition on starting with larger amounts. *See* Ex. 1030, 1 (editor’s summary); Ex. 1225 ¶ 105. And although Forshew tested the TAm-Seq method using DNA equivalent to that present in from 30 to 150 microliters of blood plasma, Forshew actually collected that blood as multiple one-milliliter (1000 microliter) aliquots. *See* Ex. 1225 ¶ 105 (citing Ex. 1030, 10). We further find that “obtaining additional plasma [and, thus, even larger amounts of starting DNA] would have involved nothing more than a routine blood draw.” *See id.*

We further note that the 3,600 ng of DNA used in Wagel’s hybridization step was not native DNA extracted from a single tumor. *See* Ex. 1033, 9 (Methods). Rather, and as explained by Dr. Quackenbush, DNA from 12 tumor lines was extracted from paraffin blocks, PCR amplified, and pooled before being exposed to Wagel’s hybridization capture probes. Ex. 1225 ¶¶ 106–107. In this respect, Wagel states:

Massively parallel sequencing libraries (Illumina) that contain barcoded universal primers were generated with the use of genomic DNA from formalin-fixed, paraffin-embedded tumor material. After preamplification and DNA quantification, equimolar pools were generated consisting of 12 barcoded tumor DNAs. These DNA pools were subjected to solution-phase hybrid capture with biotinylated RNA baits targeting all exons from 137 actionable cancer genes. Each hybrid capture reaction was sequenced

Ex. 1033, 91; *see also, id.* (“DNA libraries were pooled by mixing 300 ng of each library”). As such, we understand Wagel’s use of 3,600 ng to be the result of attempting to simultaneously survey DNA amplified from twelve different tumors and identified by barcodes, and suggests that substantially less DNA would be required to analyze DNA from a single tumor as embodied by Forshew. *See* Ex. 1225 ¶ 107.

In view of the above, we are not persuaded by Patent Owner’s argument that there was no motivation to combine Forshew and Wagel because of differences in the amount of DNA used in each method.¹⁹

b) Substituting or Combining PCR with Biotinylated Baits

Relying on Mamanova, Patent Owner argues that “a POSA would have needed to evaluate seven parameters associated with the disadvantages of capture probes before discarding the core reaction principle of Forshew’s PCR primer-based assay and replacing it with [hybridization-based biotinylated baits].” POR 72–73 (citing Ex. 1007, 111, Tab 1; Ex. 2031 ¶¶ 211–213). Patent Owner similarly points to Mertes as evidence that “PCR

¹⁹ We also do not find persuasive, Patent Owner’s subsidiary arguments, including that Forshew recommends against using Wagel’s capture probes for plasma samples, and “criticizes capture probes for being ‘expensive,’” for the reasons set forth in Section III.C, pages 22–23, of the Reply, which we adopt in full. *See* POR 71–72, 75–76; Sur-reply 21–22.

primers and [hybridization-based biotinylated baits] have different ‘operational characteristics’ that ‘naturally vary in their suitability for different fields of application.’” *Id.* at 73 (citing Ex. 1006, 373, 378). Considering the differences in, e.g., specificity, uniformity, reproducibility, and sensitivity between the two approaches, as well as “the risks associated with modifying Forshew’s already working assay,” Patent Owner contends that one of ordinary skill in the art “would not have been motivated to make such a drastic change to Forshew’s core reaction principle.” *Id.* at 73–74; Sur-reply 17–19. Patent Owner further argues that Petitioner has failed to articulate with particularity how one of ordinary skill in the art would have modified Forshew “to include both PCR primers and capture probes.” POR 74–76; Sur-reply 22–23.

We do not find Patent Owner’s arguments persuasive for the reasons set forth at pages 20–24 of the Reply, which we adopt. We credit, in particular, Dr. Quackenbush’s testimony that,

A POSA would have been motivated to combine Forshew and Wagle by using Wagle’s solution-phase hybridization RNA baits as capture probes, either in place of, or in addition to, Forshew’s PCR primer capture probes. By the priority date, a POSA would have considered these amplification techniques to be relatively interchangeable, as discussed in Section V.C.3,^[20] which I incorporate by reference here. Indeed, Forshew cites Wagle. Ex. 1030 at 11 (citation 35). And Wagle suggests that in certain circumstances, PCR-capture based techniques may be less optimal (although not ineffective), and thus proposes its exon capture approach as a related technique suitable for the capture portion of the general workflow.

²⁰ Ex. 1020 ¶¶ 68–80.

Ex. 1020 ¶ 180 (internal footnote omitted). We further note that to show obviousness, Petitioner need not show how to bodily incorporate Wagel’s hybridization capture probes into Forshew’s PCR-based method. *See In re Keller*, 642 F.2d 413, 425 (C.C.P.A. 1981). “Rather, the test is what the combined teachings of the references would have suggested to those of ordinary skill in the art.” *Id.* In this respect, and in addition to Dr. Quackenbush’s testimony, we note that Wagel itself employs PCR amplification in combination with hybridization capture probes, which weighs in favor of the asserted combination. *See* Ex. 1033, 91; Ex. 1225 ¶¶ 106–107. Considering all the arguments and evidence of record, we find that Petitioner has shown sufficiently that one of ordinary skill in the art would have been motivated to combine the teachings of Forshew and Wagel so as to arrive at the claim invention.

4. Objective Indicia of Non-Obviousness

Objective indicia of non-obviousness may include long-felt but unsolved need, failure of others, unexpected results, commercial success, copying, licensing, industry praise, and expert skepticism. *Mintz v. Dietz & Watson, Inc.*, 679 F.3d 1372, 1379 (Fed. Cir. 2012). “[O]bjective indicia ‘may often be the most probative and cogent evidence of nonobviousness in the record,’” and “help turn back the clock and place the claims in the context that led to their invention.” *Id.* at 1378. Evidence of objective indicia of non-obviousness “must always when present be considered en route to a determination of obviousness.” *Transocean Offshore Deepwater Drilling, Inc. v. Maersk Drilling USA, Inc.*, 699 F.3d 1340, 1349 (Fed. Cir. 2012); *see also Apple Inc. v. Samsung Elecs. Co.*, 839 F.3d 1034, 1048 (Fed. Cir. 2016) (en banc). Such evidence, however, does not necessarily control the

obviousness conclusion. *See, e.g., Pfizer, Inc. v. Apotex, Inc.* 480 F.3d 1348, 1372 (Fed. Cir. 2007). (“Here, the record establishes such a strong case of obviousness that Pfizer’s alleged unexpectedly superior results are ultimately insufficient.”).

But objective indicia of non-obviousness are “only relevant to the obviousness inquiry ‘if there is a nexus between the claimed invention and the [objective indicia of non-obviousness].’” *In re Affinity Labs of Tex., LLC*, 856 F.3d 883, 901 (Fed. Cir. 2017) (quoting *Ormco Corp. v. Align Tech., Inc.*, 463 F.3d 1299, 1312 (Fed. Cir. 2006)). For objective indicia of non-obviousness to be accorded substantial weight, their proponent must establish a nexus between the evidence and the merits of the claimed invention. *ClassCo, Inc. v. Apple, Inc.*, 838 F.3d 1214, 1220 (Fed. Cir. 2016). “[A] patentee is entitled to a rebuttable presumption of nexus between the asserted evidence of secondary considerations and a patent claim if the patentee shows that the asserted evidence is tied to a specific product and that the product ‘is the invention disclosed and claimed.’” *Fox Factory, Inc. v. SRAM, LLC*, 944 F.3d 1366, 1373 (Fed. Cir. 2019) (quoting *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1392 (Fed. Cir. 1988)).

Relying on paragraph 228 of Dr. Furneaux’s Declaration (Ex. 2031), Patent Owner contends it is “entitled to a presumption of nexus because Personalis’ NeXT Personal[®] assay embodies claim 1 of the ’033 patent.” POR 80; *see* Sur-reply 26. Patent Owner further asserts—without citation or elaboration—that “the evidence independently shows a nexus.” POR 80. Patent Owner further contends that its Personalis’ NeXT Personal assay has “received widespread industry attention and praise due in part to its

sensitivity,” as evidenced by its citation as a Top 10 Innovation of 2022 in *The Scientist*, and one-page reports in *Precision Online Medicine* and *genomeweb*, regarding the adoption of Personalis’ NeXT Personal in a UK TRACERx study, and “a collaboration with AstraZeneca to explore the cancer sequencing firm’s NeXT Personal molecular residual disease (MRD) test in clinical research and drug development,” respectively. *Id.* at 80–91 (citing Ex. 2001; Ex. 2153; Ex. 2154 (discussing initial results of the AstraZeneca collaboration); Ex. 2156; Ex. 2035 ¶¶ 11, 14–20). According to Patent Owner, “industry[] experts held skepticism toward solving the challenges of conducting highly sensitive genome sequencing using capture probes” and the difficulties in “‘how to handle the massive amount of sequencing data’ when utilizing NGS for profiling cancer genomes.” *Id.* at 82–83 (citing Ex. 1030, 10; Ex. 1031; Ex. 1004, 48; Ex. 2031 ¶¶ 231–232; Ex. 1017, 3).

Paragraph 228 of Dr. Furneaux’s Declaration, the only substantive evidence Patent Owner provides in support of nexus, is a bare statement that “Personalis is entitled to a presumption of nexus because, as detailed in the following chart, Personalis’ NeXT Personal® assay embodies claim 1 of the ’033 Patent.” Ex. 2031 ¶ 228. The referenced chart appears to compare claim 1 of the ’033 patent to cut-and-paste excerpts of Exhibit 2008, an attorney-generated claim chart submitted with Patent Owner’s Preliminary Response. *Id.*

In our Institution Decision, we considered Petitioner’s argument that “Patent Owner’s nexus evidence is supported by ‘only a cursory attorney-created claim chart,’” and invited the parties to more fully “present argument and evidence on this issue,” such that we could “consider objective evidence

of non-obviousness after development of a full trial record.” DI 35. Patent Owner’s mere inclusion of excerpts from the same cursory claim chart in Dr. Furneaux’s Declaration—absent any substantive investigation or guidance by Dr. Furneaux—and, moreover, absent any substantive explanation or argument in Patent Owner’s briefs, does not suffice. *See* Ex. 1223, 35:15–38:24 (Dr. Furneaux’s testimony that his input in reviewing the claim chart was merely “to check that what’s in the left-hand-side column there is, in fact, recapitulated in the right-hand-side column.”). Patent Owner has thus, “left it up to us to figure out” whether nexus exists. *See* Reply 26 (citing *Microsoft Corp. v. Syncloud Technologies, LLC*, No. 2020-00316 (PTAB Jun. 14, 2021)). We decline to do so. *See DeSilva v. DiLeonardi*, 181 F.3d 865, 867 (7th Cir. 1999) (“A brief must make all arguments accessible to the judges, rather than ask them to play archaeologist with the record.”). Considering the still incomplete record before us, Patent Owner has not satisfied its threshold burden of showing that its asserted objective indicia of non-obviousness is tied to a specific product, where that product “is the invention disclosed and claimed.” *See Fox Factory*, 944 F.3d at 1373.

A *prima facie* case of nexus cannot be made by simply showing that “the patent claims broadly cover the product that is the subject of the evidence of secondary considerations.” *See Fox Factory*, 944 F.3d at 1377. Accordingly, even were we to accept Dr. Furneaux’s conclusory testimony that claim 1 “embodies” (i.e., covers) the NeXT Personal assay, that is not alone sufficient to establish coextensiveness and the necessary degree of correspondence between the product and scope of the claims entitling Patent Owner here to a presumption of nexus. Dr. Furneaux does not, for example,

explain why alleged success of the NeXT Personal assay is even substantially due to “the invention disclosed and claimed” as opposed to, for example, some significant unclaimed features. *See id.* at 944 F.3d at 1373; *see also* Ex. 1225 ¶¶ 108–111 (Dr. Quackenbush disputing the accuracy and relevance of Dr. Furneaux’s sensitivity arguments).

In view of the above, we find that Patent Owner has not met its burden on the nexus issue. *WMS Gaming Inc. v. Int’l Game Tech.*, 184 F.3d 1339, 1359 (Fed. Cir. 1999) (“The patentee bears the burden of showing that a nexus exists.”).

We further note that, “[w]here the offered secondary consideration actually results from something other than what is both claimed and *novel* in the claim, there is no nexus to the merits of the claimed invention.” *In re Kao*, 639 F.3d 1057 (Fed. Cir. 2011). In this respect, Patent Owner anchors its evidence of industry praise and skepticism largely on the alleged sensitivity of its commercial product. POR 82–83 (e.g., referencing “the challenges of conducting highly sensitive genome sequencing using capture probes” and “difficulties regarding ‘how to handle the massive amount of sequencing data’”). But the sensitivity allegedly displayed by Patent Owner’s commercial product is not claimed, or even discussed in the Specification. Nor does Patent Owner link any patent claim or portion of the Specification to a solution for handl[ing] . . . “massive amount(s) of sequencing data.” *See* POR 84 (citing Ex. 1017, 3). Moreover, to the extent Patent Owner’s alleged high sensitivity derives from the use of “genome sequencing using capture probes,” such techniques would appear to be encompassed by the prior art under our construction of “capture probes.”

Considering the above, we ascribe no weight to Patent Owner’s evidence of objective indicia of non-obviousness.

5. Conclusion as to Ground 2, Claims 1–23

Considering the record before us, and in light of our construction of “capture probes,” we find that Petitioner has shown persuasively that claims 1–23 are unpatentable in view of Forsheew and Wagle.

G. Ground 3: Obviousness in View of Wagle and Chan

As to Ground 3, Petitioner challenges claims 1–9, 11–23 as unpatentable for obviousness in view of Wagle, and Chan. *See* Pet. 16, 51–80; Ex. 1020 ¶¶ 191–265; Ex. 1225 ¶¶ 104–107. Patent Owner opposes. POR 76–79; Sur-reply 23–26. Ex. 2031 ¶¶ 219–226. As Wagle has been discussed above, we begin with an overview of Chan.

1. Overview of Chan (Exhibit 1008)

Chan discloses “shotgun massively parallel sequencing (MPS) of plasma DNA from cancer patients to scan a cancer genome noninvasively.” Ex. 1008, 211. According to Chan, “[t]he presence of tumor-derived DNA in the plasma of cancer patients offers exciting opportunities for the detection and monitoring of cancer.” *Id.* at 211. Chan states, however, that, “[m]ost of the previously published work on plasma DNA as a cancer marker has focused on the detection of specific and predetermined molecular targets known to be associated with cancer by means of such methods such as the PCR.” *Id.*; *see also id.* at 223 (noting that “solution-phase hybridization-based capture approaches have already been successfully used to enrich plasma DNA for noninvasive prenatal diagnosis” (citing Liao, Ex. 1009)). In contrast to this targeted approach, Chan employed a “nontargeted random—or

shotgun–sequencing approach” “to obtain a noninvasive, genomewide view of cancer associated copy number variations and mutations in DNA in plasma.” *Id.* at 212.

Illustrating the shotgun MPS approach, Chan follows the proportions of cancer-associated chromosomal aberrations in the plasma of four hepatocellular carcinoma patients over time, before and after resection. *Id.* at 211 (“Four hepatocellular carcinoma patients . . . were recruited. DNA was extracted from the tumor tissues, and the preoperative and postoperative plasma samples of these patients were analyzed with shotgun MPS.”), 212, 217–219 (“MPS data from the tumor DNA and the buffy coat DNA from each of the 4 HCC patients were compared”); *see also id.* at 219–221 (MPS analysis on patient with breast and ovarian cancers). Chan concludes that shotgun MPS

would allow genomic aberrations harbored by tumor cells within a patient to be scanned at different levels of resolution, ranging from a global view of cancer-associated copy number aberrations to point mutations carried by different clones of tumor cells. This approach would allow qualitative and quantitative aspects of such aberrations to be followed serially, as evidenced by the changes we observed in the pre- and postsurgery plasma samples.

Id. at 221.

Chan further states that, although “the shotgun sequencing approach we have described is relatively expensive compared with many targeted approaches . . . cost is not expected to be as much of an obstacle in the near future.” *Id.* at 221. According to Chan,

[o]ne can also envision the possibility of using shotgun sequencing of a plasma sample upon presentation of a cancer patient, mining the shotgun sequencing data of plasma DNA for copy number aberrations and SNVs, and then specifically

targeting such sequences for serial monitoring purposes. In this regard, solution-phase hybridization-based capture approaches have already been successfully used to enrich plasma DNA for noninvasive prenatal diagnosis (34, 35). Hence, we see that the targeted and shotgun approaches can be used in combination for cancer detection and monitoring.

Id.

2. Analysis of Ground 3

Chan “teaches using solution-phase hybridization-based capture probes that target particular polymorphisms identified in an initial whole-genome sequencing assay to serially monitor and detect cancer in a patient’s plasma over time.” *See* Ex. 1008. 211 (discussing genomic changes observed in pre- and postsurgery plasma samples); Pet. 53; Ex. 1020 ¶¶ 195–197.

Petitioner points to Chan as disclosing step 1(d), and admits that Wagle does not expressly disclose repeating claim steps 1b and 1c as required by this claim step. *See* Pet. 67–68. According to Petitioner and Dr. Quackenbush, Chan would have motivated one of ordinary skill in the art to combine its teachings with “Wagle to employ solution-phase hybridization probes, including solution-phase hybridization probes targeting additional types of polymorphisms, followed by NGS, and repeating the procedure at additional time points to monitor tumors in patients.” Pet. 53; Ex. 1020 ¶ 197.

Patent Owner, however, contends that Ground 3 fails because “Wagle in view of Chan does not teach or suggest limitation 1(d).” POR 77–79; Sur-reply 23–26. Claim step 1(d) recites, “repeating steps (b)–(c) on a subsequently obtained biological sample from said individual” which, as noted in our Institution Decision, encompasses serial monitoring of a patient under treatment for cancer. *See* DI 38. Patent Owner argues that one of ordinary skill in the art “would not have been motivated to repeat Wagle’s

solution-based exon capture technique on . . . ‘subsequently obtained tumor samples,’” as this entails “an undesirable invasive procedure.” POR 77–78 (citing Ex. 1030, 1 (Forshe’s statement that “[b]iopsies are invasive and costly and only provide a snapshot of mutations present at a given time and location.”); Ex. 2111, 2; Ex. 1033, 90; Ex. 2031 ¶¶ 222–225); Sur-reply 23–24. Nor, Patent Owner argues, would there have been motivation to repeat Wagle’s assay using “a plasma sample that included cell-free tumor DNA” because “cell-free tumor DNA is not present in sufficient quantities to have motivated” its use with a reasonable expectation of success.” POR 78–79 (citing Ex. 1030, 9; Ex. 2031 ¶ 226); Sur-reply 24–26.

Patent Owner’s arguments are unpersuasive for the reasons set forth at pages 24–26 of the Reply, which we adopt.

We also do not find credible Patent Owner’s argument that motivation to monitor the progression of a patient’s cancer is abrogated by the expense or invasiveness of a second or subsequent tumor biopsy even in the case of solid tumors. We also note that the “nucleic acid sample obtained from an individual” in claim 1 need not be derived from a solid tumor. Indeed, claims 10 and 11 specify that the nucleic acid sample may be obtained from “a body fluid, cell, skin, tissue, organ, or . . . blood plasma.” As such, the claimed tissue sample encompasses blood cell tumors such as the acute myeloid leukemia cells isolated from the white blood cell-enriched buffy coat fraction as taught by Ley (Ex. 1012),²¹ asserted in related, co-pending IPR2023-00224. Where the claimed tissue sample comprises blood cell tumors, any “second or subsequent tumor biopsy” entails an inexpensive and

²¹ Ley *et al.*, “DNA sequencing of a cytogenetically normal acute myeloid leukaemia genome,” 456 *Nature* 66–72 (2008). Ex. 1012.

minimally-invasive blood draw. *See generally* Ex. 1020 ¶¶ 82–84, 93; *see also* Ex. 1225 ¶ 98 (noting that Chan’s studies employed “matched tumor tissue from hepatocellular carcinoma (HCC) and normal buffy coat DNA”); Ex. 1008 at 213 (“DNA samples extracted from the buffy coat and the tumor tissues of the HCC patients were analyzed with the SurePrint G3 Human High Resolution Microarray Kit (Agilent) as previously described,”), 218 (“MPS data from the tumor DNA and the buffy coat DNA from each of the 4 HCC patients were compared”).

In view of the above, we are unpersuaded by Patent Owner’s “invasive and costly biopsy” argument.

Moreover, as discussed in Section II.F, above, and exemplified by Wagle, one of ordinary skill in the art understood that small starting amounts of DNA could be amplified by PCR prior to contact with hybridization capture probes. *See* Ex. 1033, 91 (“After preamplification and DNA quantification, equimolar pools were generated consisting of 12 barcoded tumor DNAs. These DNA pools were subjected to solution-phase hybrid capture with biotinylated RNA baits.”). Further, and as noted by Dr. Quackenbush, Chan expressly states that “***solution-phase hybridization based capture approaches have already been successfully used to enrich plasma DNA for noninvasive prenatal diagnosis***,” thus making clear that one of ordinary skill in the art would not have found the amount of DNA in patient plasma to be a limiting factor for solution-based hybridization based capture analyses. Ex. 1020 ¶ 213 (quoting Ex. 1088, 223); *see also* Pet. 63–64 (similar). As such, we are not persuaded by Patent Owner’s “insufficient DNA” argument.

As to motivation generally, although Petitioner admits that at least Wagle does not expressly disclose repeating claim steps 1b and 1c as required by claim step 1d (Pet. 67), this element is disclosed or implicit in Chan's statements that "targeted shotgun approaches can be used in combination for cancer detection *and monitoring*" (Ex. 1008, 221) (emphasis added); "[the disclosed] approach would allow qualitative and quantitative aspects of such aberrations to be *followed serially, as evidenced by the changes we observed in the pre- and postsurgery plasma samples*" (*id.*) (emphasis added); that "[t]he presence of tumor-derived DNA in the plasma of cancer patients offers exciting opportunities for the detection *and monitoring* of cancer" (*id.* at 222 (emphasis added)); and that "[o]ne can also envision the possibility of using shotgun sequencing of a plasma sample upon presentation of a cancer patient, mining the shotgun sequencing data of plasma DNA for copy number aberrations and SNVs" (*id.* at 223 (emphasis added)). *See also* Ex. 1020 ¶ 194 (Dr. Quackenbush's testimony that "the use of solution-phase hybridization followed by NGS to serially track tumor-associated polymorphisms is expressly contemplated by Chan. Ex. 1008."), ¶ 213 (similar). Moreover, in addition to Chan, we credit Dr. Quackenbush's uncontested testimony that the use of DNA marker analysis to serially monitor tumor DNA in a patient's blood plasma was well known in the art.

See, e.g., Ex. 1020 ¶ 85 (citing Ex. 1002²² (Leary); Ex. 1015²³ (McBride), Ex. 1031²⁴ (Meyerson)); ¶¶ 131, 133 (citing Ex. 1030 (Forsheew)).

In sum, the preponderance of the evidence of record supports Petitioner’s contention that one of ordinary skill in the art would have found it obvious to combine Wagle and Chan with a reasonable expectation of success in achieving all limitations of the asserted claims, including contested element 1(d).

3. Conclusion as to Ground 3, Claims 1–9 and 11–23

Considering the record before us, Petitioner has shown sufficiently that claims 1–9 and 11–23 are unpatentable for obviousness in view of Wagle and Chan.

III. CONCLUSION

Having analyzed the entirety of the record and assigning appropriate weight to the evidence cited by Petitioner and Patent Owner, and for the reasons we have set forth above, we determine that Petitioner has established, by a preponderance of the evidence, that claims 1–23 of the ’033 patent are unpatentable.

²² Leary *et al.*, “Development of Personalized Tumor Biomarkers Using Massively Parallel Sequencing,” 2(20) *Science Translational Medicine* 20ra14 (2010). Ex. 1002; Ex. 1026 (supplemental materials available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2858564/>).

²³ McBride, et al., “Use of Cancer-Specific Genomic Rearrangements to Quantify Disease Burden in Plasma from Patients with Solid Tumors,” 49(11) *Genes, Chromosomes & Cancer* 1062–1069 (2010).

²⁴ Meyerson, et. al., “Advances in understanding cancer genomes through second-generation sequencing,” 11 *Nature Reviews* 685–696 (2010).

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has shown by a preponderance of the evidence that claims 1–23 of the '033 patent are unpatentable;

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.²⁵

Claims	35 U.S.C. §	Reference(s) /Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–7, 10–23	§ 102	ForsheW	1–7, 10–23	
1–23	§ 103	ForsheW, Wagle	1–23	
1–9, 11–23	§ 103	Wagle, Chan	1–9, 11–23	
Overall Outcome			1–23	

²⁵ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

FOR PETITIONER:

Andrew E. Krause
IRELL & MANELLA LLP
akrause@irell.com

Amir Naini
NAINI P.C.
amir@nainipc.com

FOR PATENT OWNER:

Derek C. Walter
Adrian Percer
WEIL, GOTSHAL & MANGES LLP
derek.walter@weil.com
adrian.percer@weil.com